

$$f : \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$\begin{aligned} \min f(x) &= 4x_1 + x_2 + x_3 \\ 2x_1 + x_2 + 2x_3 &= 4 \\ 3x_1 + 3x_2 + x_3 &= 3 \\ -x_1 + x_2 - 3x_3 &= -5 \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

Metodo delle due fasi:

Fase I

$$f(x) = a_1 + a_2 + a_3$$

Calcolo dei costi:

$$c_i = c_N - c_B^T \cdot B_N^{-1}$$

$$c_1 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = -6$$

$$c_2 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix} = -3$$

$$c_3 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} = -6$$

$$c_4 = 1 - (1; 1; 1) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$c_5 = 1 - (1; 1; 1) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0$$

$$c_6 = 1 - (1; 1; 1) \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$c_7 = 0 - (1; 1; 1) \cdot \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix} = -12$$

Metodo del simplesso:

	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	2	1	2	1	0	0	4
a_2	3	3	1	0	1	0	3
a_3	1	-1	3	0	0	1	5
	-6	-3	-6	0	0	0	-12

Entra x_1 ed esce a_2 :

	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	2	1	2	1	0	0	4
a_2	3	3	1	0	1	0	3
a_3	1	-1	3	0	0	1	5
	-6	-3	-6	0	0	0	-12

Forma standard

$$\begin{aligned} \min f(x) &= 4x_1 + x_2 + x_3 \\ 2x_1 + x_2 + 2x_3 + a_1 &= 4 \\ 3x_1 + 3x_2 + x_3 + a_2 &= 3 \\ x_1 - x_2 + 3x_3 + a_3 &= 5 \\ x_1, x_2, x_3, a_1, a_2, a_3 &\geq 0 \end{aligned}$$

$$R2 \leftarrow R2 \cdot \frac{1}{3}$$

	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	2	1	2	1	0	0	4
a_2	1	1	$\frac{1}{3}$	0	$\frac{1}{3}$	0	1
a_3	1	-1	3	0	0	1	5
	-6	-3	-6	0	0	0	-12

$$R3 \leftarrow R3 - R2$$

$$\begin{array}{cccccc|c} 1 & -1 & 3 & 0 & 0 & 1 & 5 \\ 1 & 1 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 1 \\ \hline 0 & -2 & \frac{8}{3} & 0 & -\frac{1}{3} & 1 & 4 \end{array}$$

↓

	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	2	1	2	1	0	0	4
a_2	1	1	$\frac{1}{3}$	0	$\frac{1}{3}$	0	1
a_3	0	-2	$\frac{8}{3}$	0	$-\frac{1}{3}$	1	4
	-6	-3	-6	0	0	0	-12

$$R1 \leftarrow R1 - R2 \cdot 2$$

$$\begin{array}{cccccc|c} 2 & 1 & 2 & 1 & 0 & 0 & 4 \\ 2 & 2 & \frac{2}{3} & 0 & \frac{2}{3} & 0 & 2 \\ \hline 0 & -1 & \frac{4}{3} & 1 & -\frac{2}{3} & 0 & 2 \end{array}$$

↓

	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	0	-1	$\frac{4}{3}$	1	$-\frac{2}{3}$	0	2
a_2	1	1	$\frac{1}{3}$	0	$\frac{1}{3}$	0	1
a_3	0	-2	$\frac{8}{3}$	0	$-\frac{1}{3}$	1	4
	-6	-3	-6	0	0	0	-12

$$R4 \leftarrow R4 + R2 \cdot 6$$

$$\begin{array}{cccccc|c} -6 & -3 & -6 & 0 & 0 & 0 & -12 \\ 6 & 6 & 2 & 0 & 2 & 0 & 6 \\ \hline 0 & 3 & -4 & 0 & 2 & 0 & -6 \end{array}$$

↓

	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	0	-1	$\frac{4}{3}$	1	$-\frac{2}{3}$	0	2
x_1	1	1	$\frac{1}{3}$	0	$\frac{1}{3}$	0	1
a_3	0	-2	$\frac{8}{3}$	0	$-\frac{1}{3}$	1	4
	0	3	-4	0	2	0	-6

Entra x_3 ed esce a_1 :

	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	0	-1	$\frac{4}{3}$	1	$-\frac{2}{3}$	0	2
x_1	1	1	$\frac{1}{3}$	0	$\frac{1}{3}$	0	1
a_3	0	-2	$\frac{8}{3}$	0	$-\frac{1}{3}$	1	4
	0	3	-4	0	2	0	-6

$$R1 \leftarrow R1 \cdot \frac{3}{4}$$

	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	0	$-\frac{3}{4}$	1	$\frac{3}{4}$	$-\frac{1}{2}$	0	$\frac{3}{2}$
x_1	1	1	$\frac{1}{3}$	0	$\frac{1}{3}$	0	1
a_3	0	-2	$\frac{8}{3}$	0	$-\frac{1}{3}$	1	4
	0	3	-4	0	2	0	-6

$$R2 \leftarrow R2 - R1 \cdot \frac{1}{3}$$

$$\begin{array}{cccccc|c} 1 & 1 & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 1 \\ 0 & -\frac{1}{4} & \frac{1}{3} & \frac{1}{4} & -\frac{1}{6} & 0 & \frac{1}{2} \\ \hline 1 & \frac{5}{4} & 0 & -\frac{1}{4} & \frac{1}{2} & 0 & \frac{1}{2} \end{array}$$

↓

	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	0	$-\frac{3}{4}$	1	$\frac{3}{4}$	$-\frac{1}{2}$	0	$\frac{3}{2}$
x_1	1	$\frac{5}{4}$	0	$-\frac{1}{4}$	$\frac{1}{2}$	0	$\frac{1}{2}$
a_3	0	-2	$\frac{8}{3}$	0	$-\frac{1}{3}$	1	4
	0	3	-4	0	2	0	-6

$$R3 \leftarrow R3 - R1 \cdot \frac{8}{3}$$

$$\begin{array}{cccccc|c} 0 & -2 & \frac{8}{3} & 0 & -\frac{1}{3} & 1 & 4 \\ 0 & -2 & \frac{8}{3} & 2 & -\frac{4}{3} & 0 & 4 \\ \hline 0 & 0 & 0 & -2 & 1 & 1 & 0 \end{array}$$

↓

	x_1	x_2	x_3	a_1	a_2	a_3	b
a_1	0	$-\frac{3}{4}$	1	$\frac{3}{4}$	$-\frac{1}{2}$	0	$\frac{3}{2}$
x_1	1	$\frac{5}{4}$	0	$-\frac{1}{4}$	$\frac{1}{2}$	0	$\frac{1}{2}$
a_3	0	0	0	-2	1	1	0
	0	3	-4	0	2	0	-6

$$R4 \leftarrow R4 + R1 \cdot 4$$

$$\begin{array}{cccccc|c} 0 & 3 & -4 & 0 & 2 & 0 & -6 \\ 0 & -3 & 4 & 3 & -2 & 0 & 6 \\ \hline 0 & 0 & 0 & 3 & 0 & 0 & 0 \end{array}$$

↓

	x_1	x_2	x_3	a_1	a_2	a_3	b
x_3	0	$-\frac{3}{4}$	1	$\frac{3}{4}$	$-\frac{1}{2}$	0	$\frac{3}{2}$
x_1	1	$\frac{5}{4}$	0	$-\frac{1}{4}$	$\frac{1}{2}$	0	$\frac{1}{2}$
a_3	0	0	0	-2	1	1	0
	0	0	0	3	0	0	0

$z = 0$ si passa alla seconda fase.

Fase II

$$\min f(x) = 4x_1 + x_2 + x_3$$

$$-\frac{3}{4}x_2 + x_3 + \frac{3}{4}a_1 - \frac{1}{2}a_2 = \frac{3}{2}$$

$$x_1 + \frac{5}{4}x_2 - \frac{1}{4}a_1 + \frac{1}{2}a_2 = \frac{1}{2}$$

$$-2a_1 + a_2 + a_3 = 0$$

$$x_1, x_2, x_3, a_1, a_2, a_3 \geq 0$$

Si ponga $a_1, a_2, a_3 = 0$:

	x_1	x_2	x_3	a_1	a_2	a_3	b
x_3	0	$-\frac{3}{4}$	1	$\frac{3}{4}$	$-\frac{1}{2}$	0	$\frac{3}{2}$
x_1	1	$\frac{5}{4}$	0	$-\frac{1}{4}$	$\frac{1}{2}$	0	$\frac{1}{2}$
a_3	0	0	0	-2	1	1	0
	0	0	0	3	0	0	0

Calcolo dei costi della nuova tabella:

$$c_1 = 4 - (1; 4) \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0$$

$$c_2 = 1 - (1; 4) \cdot \begin{pmatrix} -\frac{3}{4} \\ \frac{5}{4} \end{pmatrix} = -\frac{13}{4}$$

$$c_3 = 1 - (1; 4) \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 0$$

$$c_4 = 0 - (1; 4) \cdot \begin{pmatrix} \frac{3}{2} \\ \frac{1}{2} \end{pmatrix} = -\frac{7}{2}$$

Tabella risultante:

	x_1	x_2	x_3	b
x_3	0	$-\frac{3}{4}$	1	$\frac{3}{4}$
x_1	1	$\frac{5}{4}$	0	$\frac{1}{2}$
	0	$-\frac{13}{4}$	0	$-\frac{7}{2}$

Entra x_2 ed esce x_1 :

	x_1	x_2	x_3	b
x_3	0	$-\frac{3}{4}$	1	$\frac{3}{4}$
x_1	1	$\frac{5}{4}$	0	$\frac{1}{2}$
	0	$-\frac{13}{4}$	0	$-\frac{7}{2}$

$$R2 \leftarrow R2 \cdot \frac{4}{5}$$

	x_1	x_2	x_3	b
x_3	0	$-\frac{3}{4}$	1	$\frac{3}{4}$
x_1	$\frac{4}{5}$	1	0	$\frac{2}{5}$
	0	$-\frac{13}{4}$	0	$-\frac{7}{2}$

$$R1 \leftarrow R1 + R2 \cdot \frac{3}{4}$$

$$\begin{array}{ccc|c} 0 & -\frac{3}{4} & 1 & \frac{3}{4} \\ \frac{3}{5} & \frac{4}{5} & 0 & \frac{10}{5} \\ \hline \frac{13}{5} & 0 & 1 & \frac{9}{5} \end{array}$$

↓

	x_1	x_2	x_3	b
x_3	$\frac{3}{5}$	0	1	$\frac{9}{5}$
x_1	$\frac{4}{5}$	1	0	$\frac{2}{5}$
	0	$-\frac{13}{4}$	0	$-\frac{7}{2}$

$$R3 \leftarrow R3 + R2 \cdot \frac{13}{4}$$

$$\begin{array}{ccc|c} 0 & -\frac{13}{4} & 0 & -\frac{7}{2} \\ \frac{13}{5} & \frac{13}{4} & 0 & \frac{13}{5} \\ \hline \frac{13}{5} & 0 & 0 & -\frac{11}{5} \end{array}$$

↓

	x_1	x_2	x_3	b
x_3	$\frac{3}{5}$	0	1	$\frac{9}{5}$
x_2	$\frac{4}{5}$	1	0	$\frac{2}{5}$
	$\frac{13}{5}$	0	0	$-\frac{11}{5}$

$$x_1 = 0$$

$$x_2 = \frac{2}{5}$$

$$x_3 = \frac{9}{5}$$

$$\min f(x) = 4x_1 + x_2 + x_3 \rightarrow f(x) = 4(0) + \left(\frac{9}{5}\right) + \left(\frac{2}{5}\right) = \boxed{\frac{11}{5}}$$